

Practice Quiz 04D — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 04

Set identities by logical laws; unions of families and power sets.

Complements are taken in a fixed universal set. Name a definition or law at every step of part (a).

1. (a) Prove $(A \cap B) \cup (A \cap B^c) = A$ by a chain of equivalences beginning with membership in the left side.

Solution For an arbitrary object x , the definitions give

$$x \in (A \cap B) \cup (A \cap B^c) \iff (x \in A \wedge x \in B) \vee (x \in A \wedge x \in B^c).$$

By the distributive law and the definition of complement, this is equivalent to

$$x \in A \wedge (x \in B \vee x \notin B).$$

The parenthesized statement is true by excluded middle, so the identity law reduces the conjunction to $x \in A$. Therefore the sets are equal. ■

- (b) Let $\mathcal{F} = \{\{p\}, \{q\}\}$. Compute $\bigcup \mathcal{F}$ and then $\mathcal{P}(\bigcup \mathcal{F})$.

Solution $\bigcup \mathcal{F} = \{p, q\}$, so

$$\mathcal{P}(\bigcup \mathcal{F}) = \{\emptyset, \{p\}, \{q\}, \{p, q\}\}. \quad \blacksquare$$